Bangladesh Mathematics and Science Olympiad

BdMSO Sample Questions

Mathematics

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Organizer
Society for the Popularization of Science, Bangladesh (SPSB)
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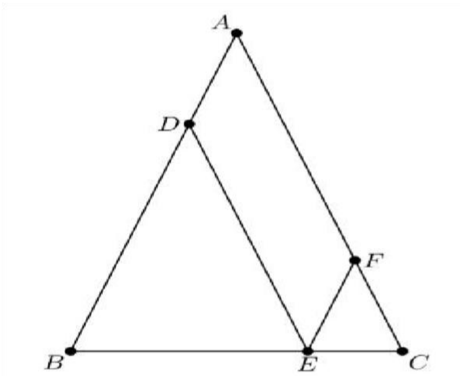
Short Answer Problems		
#	**Question**	**Answer**
1	In $\triangle ABC$, $AB = AC = 28$ and $BC = 20$. Points $D, E,$ and F are on sides $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ respectively, such that $\overline{DE}$ and $\overline{EF}$ are parallel to $\overline{AC}$ and $\overline{AB}$ respectively. What is the perimeter of parallelogram ? 	
2	How many nonnegative integers can be written in the form $a_7 \cdot 3^7 + a_6 \cdot 3^6 + a_5 \cdot 3^5 + a_4 \cdot 3^4 + a_3 \cdot 3^3 + a_2 \cdot 3^2 + a_1 \cdot 3^1 + a_0 \cdot 3^0$, where $a_i \in \{-1, 0, 1\}$ for $0 \leq i \leq 7$?	
3	Cards are numbered 1,2,...,300. What is the maximum size of a subset S of cards such that no two distinct elements of S sum to 401?	
4	Using digits {1,3,5,7,9}, find the sum of all 5-digit odd numbers formed?	
5	Find the remainder when the product P is divided by 1000. P=1^1 * 2^2 * 3^3 * ... * 2025^2025	
6	A merchant had some oranges. He sold half of them and 10 more, then half of the rest and 20 more, then half of what was left and 30 more and finally he had 40 oranges left. How many oranges did he have at the beginning?	
7	There are 100 passengers on a train, each paying $7. They only have 10,15 and 20$ bills. Trading among themselves they all pay correctly and get the right change. What is the smallest possible number of bills used?	

Essay Problems	
1	A hour digital clock shows times from 1:00 to 12:59 (then repeats). How many times in a 24-hour period is the sum of the digits of the hours equal to the sum of the digits of the minutes?
Ans. 1	
2	You have the digit string 12121 (five digits in that order). You must insert exactly one plus sign (+) and zero or more multiplication signs (x) between the digits to form valid arithmetic expressions produce an odd integer result?
Ans. 2	

Exploration Problems

1.

A set of 13 integers is arranged in nondecreasing order. Let the median be mm.

i. The mean is m+2.

ii. The mode is 2m2m, occurring exactly 4 times and uniquely the most frequent value.

iii. No other integer appears 4 or more times.

Find mm and construct one valid distribution.

2.

A palindrome number is a positive integer that can be read the same way in either direction. For instance, 1001, 3443 and 91219 are palindrome numbers. Find all 5-digit palindrome numbers divisible by 44.

3.

Cyclic quadrilateral ABCD has lengths BC=CD=3 and DA=5 with ∠CDA=120∘. What is the length of the shorter diagonal of ABCD?